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Self-supervised deep learning for detection of forest disturbance types in a subtropical ecosystem using transformer and Sentinel-1 and Sentinel-2 time series data

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ABSTRACT

Accurate and timely detection of forest disturbance types is crucial for evaluating ecosystem health and global climate stability. Time series remote-sensing data offers valuable spatiotemporal information. However, frequent cloud cover in subtropical regions disrupts the temporal consistency of optical satellite data. In addition, the impact of different data sources on modeling accuracy and the challenge of acquiring large and labeled datasets for deep learning are considerable obstacles. In this study, a novel positional encoding module was designed to handle the irregular Sentinel-2 time series. The Transformer model based on this positional encoding module effectively fused Sentinel-1 and Sentinel-2 data. Meanwhile, self-supervised learning was used to address the issue of insufficient samples. The improved Transformer model was successfully applied to detect clear-cutting and disease disturbances with a F1 score of 0.95. Our results highlighted that appropriate encoding techniques increased the model's performance by between 5% and 14%. This research also found that while Sentinel-2 data alone yields good accuracy, combining Sentinel-1 and Sentinel-2 data improved model accuracy. The self-supervised learning model achieved higher accuracy (0.95) than the supervised model (0.86) on a limited data (20% of the training samples) and exhibited stable accuracy across different training data proportions.

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Forest disturbance types; time series; Sentinel-1/2; self-supervised deep learning; Transformer

1. Introduction

Subtropical forest ecosystems play a vital role in climate regulation, ecological balance maintenance, and sustainable economic development (Hethcoat et al. 2021). However, these forests face growing threats from various disturbance factors, including logging, disease outbreaks, wildfires, and droughts (Bullock, Woodcock, and Olofsson 2020; Ye et al. 2021). These disturbances significantly impact the forests' ability to regulate climate and conserve biodiversity (Caputo et al. 2016; Curran and Trigg 2006). Local economic development is also affected (Knoke et al. 2023), with the timber industry experiencing substantial losses due to disease-induced wood quality degradation (Bastit, Brunette, and Montagné-Huck 2023). While some logging practices can provide economic benefits, illegal logging activities exacerbate forest degradation and can even trigger ecological disasters (Tang et al. 2019). Therefore, accurately identifying the type and timing of forest disturbances is crucial for ensuring forest health and promoting sustainable economic development.

Medium-resolution optical satellite imagery, such as Landsat (Shimizu et al. 2023) and Sentinel-2 (Slagter et al. 2023), has been widely used for forest disturbance detection due to its convenient acquisition schedule and extensive temporal archive. However, frequent cloud cover in subtropical regions poses a significant challenge for developing robust forest disturbance detection algorithms (Verbesselt et al. 2010; M. Zhang et al. 2024; B. Yang et al. 2022). Clouds introduce long temporal gaps in the time series data, resulting in irregular time series patterns across years and locations (Slagter et al. 2023; Verhegghen, Eva, and Achard 2015). Conversely, Synthetic Aperture Radar (SAR) data offers all-weather capabilities, making it a valuable

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alternative (R. Zhang et al. 2024). SAR is sensitive to vegetation canopy structure and growth conditions. The data with different radar wavelengths and polarization modes can produce differentiated responses for different disturbance types (Dalponte et al. 2023; Zhao et al. 2022). Sentinel-1 SAR data, in particular, holds promise for detecting small-scale and short-term disturbances due to its high revisit frequency and fine spatial resolution (Doblas et al. 2022; Sun et al. 2023; Watanabe et al. 2021; Zhao et al. 2022). However, SAR images are inherently noisy, and their interpretability can be affected by factors like terrain and surface moisture (Ballère et al. 2021). To overcome these limitations, leveraging the complementary strengths of different remote sensing data sources through data fusion is a compelling approach. This has the potential to improve the accuracy and timeliness of forest disturbance monitoring (Joshi et al. 2016). However, a deeper understanding of how different data sources influence detection performance for various disturbance types remains an area of active research. Further investigation will be crucial for developing optimal multi-source fusion methods for robust and comprehensive forest disturbance detection in challenging environments like subtropical regions.

Two primary approaches exist for obtaining the type and spatiotemporal distribution of forest disturbances: two-step attribution and one-step attribution. The two-step method first identifies the disturbance location and boundaries before classifying the disturbance type (Stahl et al. 2023). While algorithms like LandTrendr, Breaks for Additive Seasonal and Trend (BFAST), and Continuous Change Detection and Classification (CCDC) can be employed for this initial detection (Mitchell, Rosenqvist, and Mora 2017), most of these methods are adapted to a single data source and require high-quality data (Cardille et al. 2022; Morrison et al. 2018). One-step disturbance attribution offers a streamlined alternative to the two-step approach. It directly classifies forest disturbance types and their spatiotemporal distribution from time series data in a single step. Studies suggest that one-step methods achieve comparable performance to two-step methods (Schroeder et al. 2017). Despite their advantages, one-step attribution models can be more complex to develop and train compared to two-step methods as mainstream one-step methods rely on handcrafted features, which can be time-consuming and potentially subjective to extract from time series data (Schroeder et al. 2017). Additionally, they often treat time series data as independent observations, neglecting the inherent temporal dependencies within the data (Weerakody et al. 2021).

Advancements in deep learning algorithms, particularly those that can effectively capture the temporal dependencies in time series data, hold promise for improving the accuracy and robustness of one-step attribution methods (e.g. convolutional neural networks (CNNs), recurrent neural networks (RNNs), convolutional long short-term memory networks (convLSTM)) (He et al. 2024). As research progresses in deep learning, we can expect further advancements in the accuracy and interpretability of one-step attribution methods, making them even more valuable tools for forest disturbance detection using remote sensing data. Deep learning offers a powerful toolbox for analyzing remote sensing data, enabling the extraction of valuable insights into real-world changes (Ma et al. 2019). Compared to traditional machine learning methods, deep learning models excel at automatically extracting spatiotemporal features from time series imagery in an end-to-end manner, eliminating the need for laborious manual feature engineering (Yan et al. 2022). CNNs, gated RNNs, and Transformers have emerged as effective and accurate solutions for remote sensing time series modeling tasks (Ben et al. 2018; Dou et al. 2021; Huang et al. 2023; Pelletier, Webb, and Petitjean 2019; Weerakody et al. 2021; Wen et al. 2023).

However, one of the greatest challenges is its irregularity introduced by cloud cover and precipitation in optical remote sensing data (B. Yang et al. 2022). Traditional methods such as monthly compositing or missing value imputation are frequently employed to address irregularities (Marinelli et al. 2023). However, these approaches tend to obscure the temporal information associated with disturbance events and could lead to attenuation of disturbance signals. For deep learning methods, irregular time series data also pose challenges in feature extraction, often resulting in model instability and poor generalizability (M. Zhang et al. 2024). Several approaches have been explored to address the issue of irregular time series data including weighted methods based on time interval information and time feature fusion methods (Tarasiou, Chavez, and Zafeiriou 2023; B. Yang et al. 2022; M. Zhang et al. 2024). For example, Gao et al. (2019) assigned weights to observations based on the time interval information**.** These weights influence the output of the input and forget gates in Long Short-Term Memory (LSTM) networks, controlling the impact of observations from different time steps. B. Yang et al. (2022) and Zhu et al. (2017) implemented methods

that incorporate dedicated time gates specifically designed to process temporal information. These gates extract time features and fuse them with features generated by other network units.

While Transformers are superior in capturing global features of time series, they still face challenges in encoding temporal information in the presence of irregular remote sensing data (Tarasiou, Chavez, and Zafeiriou 2023). Currently, Transformer-based methods typically address time series by improving the positional encoding structure. For instance, utilizing learnable positional encodings or timestamps to replace standard Transformer positional encodings (Wen et al. 2023). The day of year (DOY) is the mostly used data to support temporal information for model. It can be used as a part of input data or as positional encoding to embed into the Transformer model directly (Huang et al. 2023; Tarasiou, Chavez, and Zafeiriou 2023; Yuan and Lin 2021). The enhancement of positional encoding strengthens the Transformer's ability to extract temporal features. Positional encoding methods designed for equal-interval sequences (e.g. text sequence, equal-interval time series) may not be suitable for irregular time series containing timestamp information. This is because equal-interval sequences maintain fixed element order and relative position relationships, they use a unified position encoding matrix across different sequences. However, irregular time series exhibit variable time information, leading to disparities among individual time series. Consequently, further investigation is warranted to design efficient and succinct approaches for constructing positional encoding adapted to irregular time series.

Deep learning has emerged as a powerful tool for forest disturbance detection, but its success hinges on the availability of large labeled datasets (Hethcoat et al. 2021). Acquiring such datasets can be a significant challenge, often requiring expensive field surveys or expert-driven visual interpretation (Xu, Ma, and Zhang 2024). The scarcity of labeled data in many regions can lead to poor generalization performance of deep learning models due to overfitting. Self-supervised learning offers a promising solution to circumvent the small sample problem (Xu, Ma, and Zhang 2024). It involves pre-training deep learning models on vast amounts of unlabeled data, followed by fine-tuning on the task-specific labeled data (Xu, Ma, and Zhang 2024; Yuan and Lin 2021; X. Yang, Zhang, and Cui 2022). By leveraging the rich information within unlabeled data, self-supervised learning models can effectively avoid overfitting even when limited labeled samples are available (Xu, Ma, and Zhang 2024). Among various pre-training techniques, contrastive learning has become a popular choice (Jung et al. 2022; H. Li et al. 2022). Contrastive learning focuses on learning generalizable features by minimizing the distance between similar samples (positive pairs) in the feature space, while maximizing the distance between dissimilar samples (negative pairs) (Chen et al. 2020). While existing contrastive learning methods using single or multiple images have shown promising results (X. Li et al. 2022), most lack strategies to explicitly exploit the temporal dimension for sample construction. This limitation hinders their direct application to time series remote sensing analysis for disturbance detection. The exploration of contrastive learning for tackling sparse-sample challenges in time series disturbance detection tasks remains an exciting, under-explored area of research.

Although Transformers excel at extracting features from temporal sequences, they still face the following challenges when applied to remote sensing time series for forest disturbance detection: (1) Irregular sampling makes it difficult to encode temporal information accurately; (2) Different disturbance types exhibited distinct spectral response mechanisms, and the impact of multi-source data on the detection accuracy of each disturbance class remained unclear; (3) Acquiring large, well-annotated datasets for forest disturbance detection was highly demanding in terms of both time and resources. To address these gaps, this study proposed a novel forest disturbance type detection method that leverages multi-source temporal data and deep learning. The main innovations include:

(1) Enhanced temporal encoding of the Transformer: a time-embedding method was designed for irregular remote sensing time series, allowing the model to more accurately capture the temporal information of image sampling.
(2) Multi-source data fusion: optical imagery and SAR data were integrated into a unified detection framework. We systematically evaluated the contribution of each data source to the detection accuracy of various disturbance types.
(3) Self-supervised pretraining: a contrastive self-supervised learning strategy was introduced to pretrain the model on unlabeled data. We compared this pretraining approach with conventional supervised learning to demonstrate its effectiveness in small-sample scenarios.

By addressing these critical aspects, our study endeavors to push the boundaries of forest disturbance detection, providing valuable insights into how deep learning methodologies can be effectively utilized to enhance environmental monitoring and management practices.

2. Materials and methodology

2.1. Study area

Due to its complex topography, diverse forest ecosystems, and frequent cloud cover, Pu'er City in southwest Yunnan Province, China, presents a significant challenge for forest disturbance detection. The study focused on this region, encompassing two Sentinel-2 tiles (47QPF and 47QQF), to develop a robust approach to disturbance detection. Pu'er is the largest city in Yunnan and boasts extensive mountainous terrain accounting for 98.3% of its total area. The subtropical monsoon climate influences the region, resulting in an average annual temperature ranging from 15°C to 20.3°C, a frost-free period exceeding 315 days per year, and annual rainfall varying from 1100 mm to 2780 mm. Pu'er possesses a critical forest area with forest coverage exceeding 67%, and is positioned as a significant forest product industry base. The region exhibits distinct zonal vegetation characteristics driven by topography and climate. The dominant forest types include monsoon evergreen broad-leaved forests (e.g. Castanopsis, Schima superba, and Betula alnoides) and Simao pine forests. Simao pine forests are predominantly pure stands, mainly distributed at elevations between 1100 and 1800 m. Additionally, eucalyptus and rubber trees are widely distributed in Pu'er and serve as the main economic plantation forest. Forest disturbances in Pu'er are predominantly attributable to human activities, primarily clear-cutting, accompanied by occurrences of forest fires, diseases, and droughts. The specific spatial coverage and corresponding altitude distribution are presented in Figure 1. To mitigate influence of non-forest area and minimize computational cost, a forest mask was applied to land cover type data derived from Sentinel-2 image using the random forest algorithm, with an overall accuracy of 95.03% and a kappa coefficient of 0.90.

2.2. Dataset

2.2.1. Forest disturbance data

Forest disturbance data were collected through field investigations in June, 2022. Three categories were established for the forest area: no-disturbance, clear-cutting, and disease. For each disturbance event, detailed geospatial information was recorded, including spatial location, boundaries, and year of occurrence. Following data collection, internal quality control procedures were implemented to ensure data integrity. Samples with missing disturbance type or time information were removed, resulting in a final

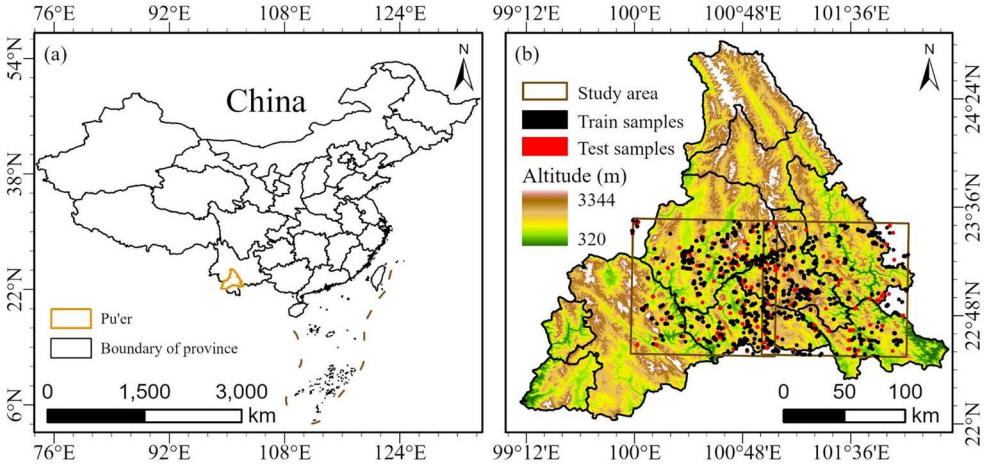


Figure 1. Location of the study area and forest disturbance samples. (a) Location of Pu'er city, (b) Study area and distribution of samples.

dataset for further analysis. Subsequently, the samples were partitioned into training and testing datasets at a ratio of 7:3. These forest disturbance polygons were converted into raster grids to facilitate integration with Sentinel-1 and Sentinel-2 time series data. The final dataset comprised rasterized disturbance polygons and time series data for model training and testing (details provided in Table 1).

2.2.2. Sentinel-1 SAR data

This study utilized Sentinel-1 data acquired in Interferometric Wide Swath (IW) mode during satellite ascents. Level-1 Ground Range Detected (GRD) data products were obtained from Google Earth Engine (GEE) (Pérez-Cutillas et al. 2023), encompassing two polarizations: Vertical-Vertical (VV) and Vertical-Horizontal (VH). Monthly mean synthesis was conducted, aggregating data over a span of 12 months annually, spanning from 2020 to 2022. Subsequently, images were exported from GEE at a resolution of 10 meters. Due to the pre-processing steps applied by the Sentinel-1 Toolbox within GEE (Veci et al. 2017), no additional pre-processing was required.

2.2.3. Sentinel-2 multispectral data

Time series of Sentinel-2 L2A level images were acquired for the period 2020-2022. Images with less than 80% cloud cover were selected. Cloud removal was performed using the cloud probability bands (pixels with probability exceeding 50% were masked out), and shadows were removed based on the Scene Classification Layer (SCL). Ten spectral bands were employed in the analysis: three visible bands (Blue, Green, Red), four red-edge bands (Red Edge 1, 2, 3, 4), one near-infrared band (Near Infrared), and two shortwave infrared bands (Shortwave Infrared 1, 2). All bands were resampled to a uniform 10-meter resolution using the nearest neighbor method. Any missing values were replaced by 0. Figure 2 illustrates the distribution of valid observation values across different Sentinel-2 tiles and years.

2.2.4. Disturbance response characteristics of different data sources

The response characteristics of multi-source data (Sentinel-1 and Sentinel-2) for different disturbance types were demonstrated using image features, spectral features, and time-series spectral features (Figures 3 and 4).

Table 1. Forest disturbance samples for model training and testing.

	Number of polygons in each year					
Disturbance type	2020	2021	2022	Area (km^2)	Training dataset (number of pixels)	Testing dataset (number of pixels)
No-disturbance	149	128	254	23.87	169554	69118
Clear-cutting	89	148	195	19.10	130258	60784
Disease	18	12	41	0.40	2598	1394

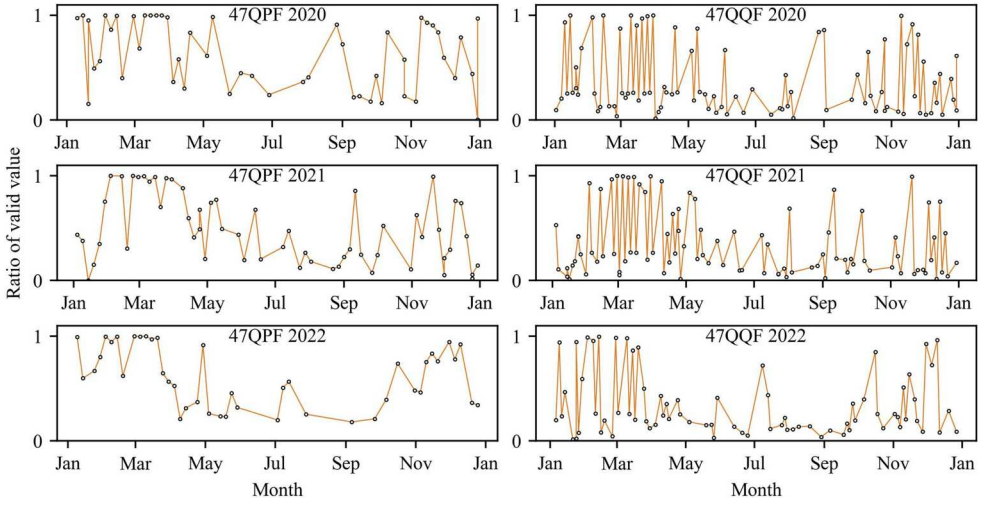


Figure 2. Ratio of valid pixels (cloud-free, shadow-free, no missing data) in Sentinel-2 tiles (2020-2022).

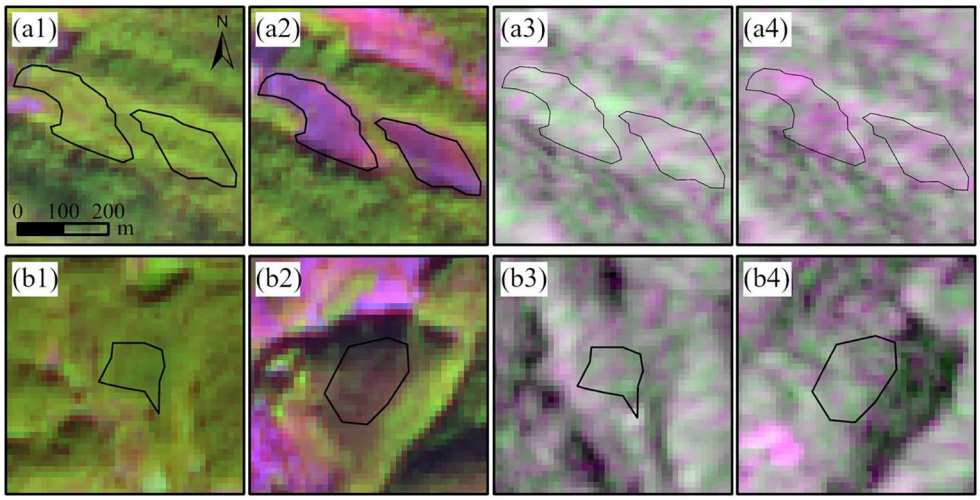


Figure 3. Image characteristics of clear-cutting and Simao pine disease for both Sentinel-1 and Sentinel-2 data. (a1) Sentinel-2 image before clear-cutting. (a2) Sentinel-2 image after clear-cutting. (a3) Sentinel-1 image before clear-cutting. (a4) Sentinel-1 image after clear-cutting. (b1) Healthy Simao pine Sentinel-2 image. (b2) Diseased Simao pine Sentinel-2 image. (b3) Healthy Simao pine Sentinel-1 image. (b4) Diseased Simao pine Sentinel-1 image.

Note: Sentinel-2 images were displayed in a false-color composite using shortwave infrared 1, near infrared, and red bands. Sentinel-1 images were displayed in combination with VV, VH, and VV.

Images from Sentinel-2 exhibited a clear response to clear-cutting events, as evident from the distinct boundaries between the cleared area and the undisturbed forest (Figure 3a1-a2). In contrast, Sentinel-1 data appeared to be heavily influenced by factors like terrain, surface roughness, and soil moisture

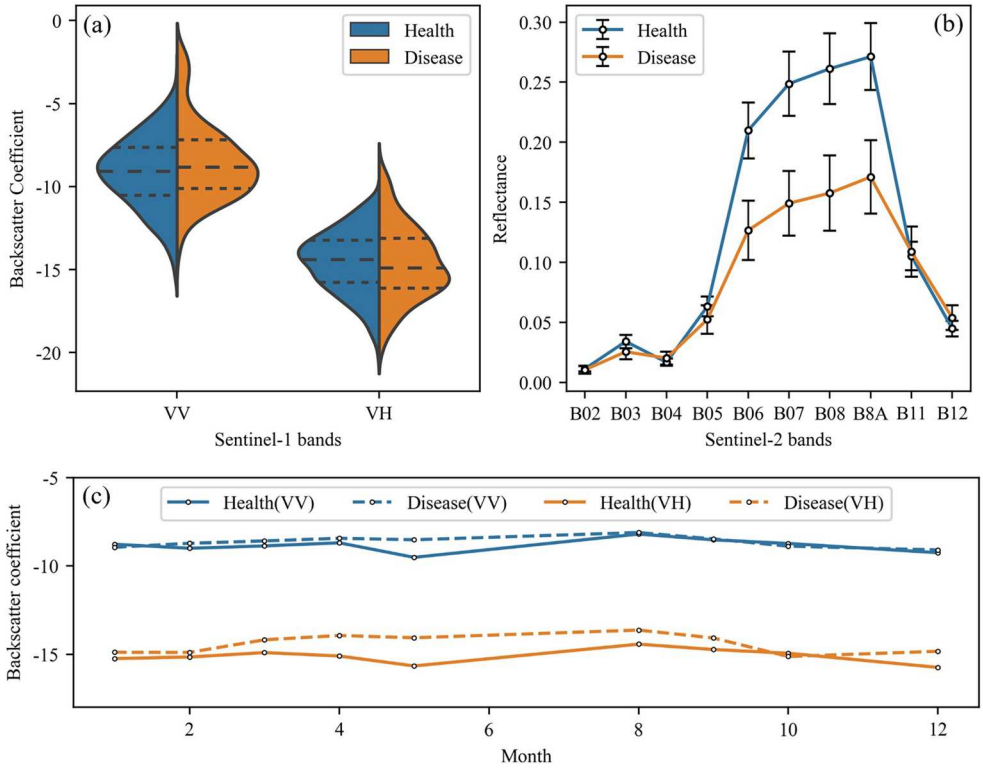


Figure 4. Spectral characteristics and time-series response of Sentinel-1 data for healthy and diseased Simao pine. (a) Spectral features of Sentinel-1. (b) Spectral features of Sentinel-2. (c) Time series of backscatter coefficient for healthy and diseased Simao pine in VV and VH polarizations.

Note: B02, blue band; B03, green band; B04, red band; B05, red edge 1 band; B06, red edge 2 band; B07, red edge 3 band; B08, near infrared band; B8A, red edge 4 band; B11, shortwave infrared 1 band; B12, shortwave infrared 2 band.

(Figure 3a3-a4). While some spatial differences could be observed between pre- and post-cutting images, they lacked clarity in defining the clear-cutting boundary. Additionally, shadows caused by topography hindered the acquisition of accurate disturbance information. Figure 3b1-b4 illustrate image characteristics of healthy and diseased Simao pine in December 2022. Unlike clear-cutting, disease in Simao pine exhibited minimal response in Sentinel-1 images, with no discernible difference between disturbed and undisturbed areas. However, Sentinel-2 images provided clearer distinction.

Figure 4 shows the spectral response differences between healthy and diseased Simao pine based on the samples displayed in Figure 3. Neither of the Sentinel-1 polarization modes (VV and VH) revealed significant spectral variations between healthy and diseased trees (Figure 4a). In contrast, Sentinel-2 data showed a more pronounced spectral difference between healthy and diseased trees, particularly in the red-edge 2, red-edge 3, near-infrared, and red-edge 4 bands (Figure 4b). Figure 4c depicted the time series curves for backscatter coefficients of healthy and diseased Simao pine in both VV and VH bands. A weak response to disease was observed in the VH band, where the backscatter coefficient of healthy pine was generally lower than that of diseased pine (except for October). There was minimal to no difference in backscatter coefficient between healthy and diseased pine in the VV band throughout the time series.

2.3. Methodology

Leveraging the standard Transformer as a foundation, this research devised the IRMST (IRregular Multi-Source Transformer) model by improving the positional encoding structure. This enhancement enabled the model to effectively capture temporal features inherent in irregular remote sensing time series data. To explore the impact of multi-source data on detection accuracy of disturbance type, three strategies for different data sources were employed, alongside supervised learning techniques, to train the IRMST model. Finally, we investigated the effectiveness of self-supervised learning in addressing challenges posed by limited training sample sizes. This investigation involved evaluating the accuracy of the IRMST model under both supervised and self-supervised learning frameworks, while considering various training sample sizes. The overall flowchart is shown in Figure 5.

2.3.1. IRMST model: IRregular multi-source time series disturbance detection with transformer architecture

This study proposed the IRMST model, which leveraged the Transformer architecture to process Sentinel-1/2 time series data for forest disturbance type detection. Transformer architecture has achieved remarkable success across a wide range of sequence modeling tasks, with its self-attention mechanism serving as the

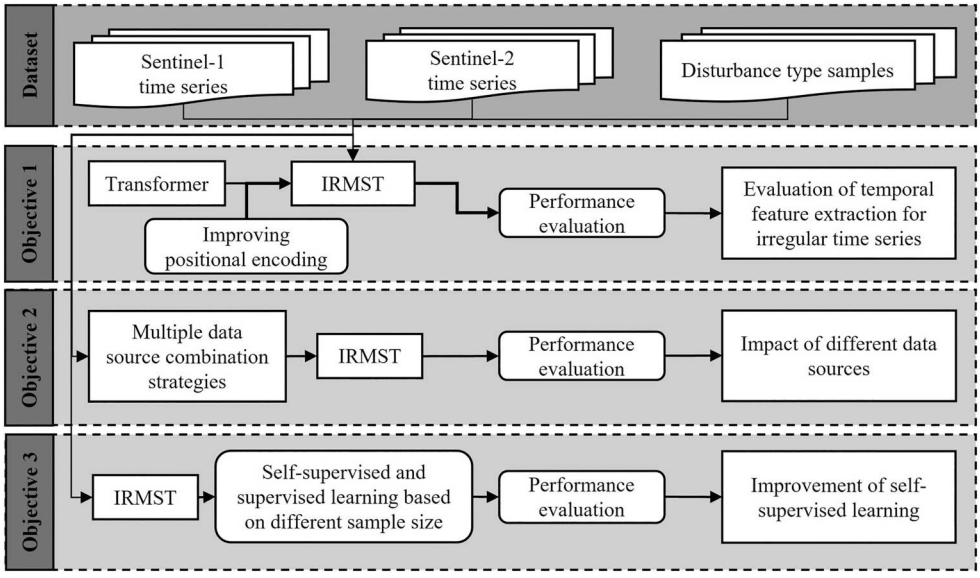


Figure 5. Flowchart of this study.

core component. Self-attention enables the Transformer to flexibly capture dependencies between any two positions in the sequence, regardless of their distance. This not only enhances the model's ability to model long-range dependencies but also allowed for highly efficient parallel computation. Before self-attention, the embedding layer maps discrete input data into continuous vector representations, providing rich semantic information and a uniform input format. This facilitatess task-specific feature learning directly from data. As self-attention is inherently unaware of sequence order, positional encoding is added to retain positional information following the embedding layer. Typically implemented via sinusoidal functions, it allows the model to perceive sequence structure without sacrificing parallelism. However, for remote sensing time series, standard embeddings and positional encodings often fall short. Such data is characterized by irregular temporal intervals, seasonality, and domain-specific temporal meanings, which fixed encodings struggle to capture. Generic embeddings also lack awareness of physical concepts like phenological stages. These challenges underscore the need for domain-adaptive encoding strategies to better model the temporal complexity of remote sensing data.

(1) Sentinel-2 multispectral time series data feature extraction

Acquiring optical remote sensing data frequently encounters challenges with cloud cover, resulting in missing values within time series images. Sentinel-2 multispectral data are susceptible to this issue as well. The observation time of each image plays a crucial role in forest disturbance detection (represented by Day of Year (DOY) in this study). Conventional Transformers utilize absolute positional encoding, thereby assigning consistent position information to all input sequences (Vaswani et al. 2017), overlooking the significant differences in observation times across diverse time series. To fully exploit the time information embedded within irregular Sentinel-2 time series, this study discarded static positional encoding and designed a new learnable encoding method.

Cloud cover and rain can lead to variations in the number of available observations and time series length for each year. Therefore, the model input was pre-defined to a constant length of 100. Missing values within shorter time series were filled with zeros. The IRMST model defined each element of the Sentinel-2 time series as a tuple with two items, encapsulating both time information (t) and observation values (v). By intricately intertwining the observation value with the observation time, the IRMST model mitigated the issue of non-uniform time distribution inherent in optical remote sensing time series. The Sentinel-2 time series is represented as:

$$T = \{ (t_i, \quad v_i) \}_{i=1}^{n} \tag{1}$$

where n denotes the length of the time series, t and v represent observed time and values, respectively. Figure 6a illustrated the module designed to handle irregular Sentinel-2 time series within the IRMST model. The time embedding layer and value embedding layer were employed to elevate the feature dimensions of temporal information and observation values, respectively. This amplification aims to enhance the model's sensitivity to subtle surface changes. Both time and value embedding were implemented through feed forward neural network. The time embedding layer mapped timestamps to vectors, while the value embedding layer mapped the reflectance values of different bands to vectors. These processes can be expressed mathematically as:

$$Embedding(x) = ReLU(W_1 x + b_1)W_2 + b_2 \tag{2}$$

where x is multi-band reflectance v_i or Day of Year (DOY) t_i. W_1 and W_2 are weight parameters of the embedding layer, and b_1 and b_2 are the bias parameters. The $ReLU$ represents the $ReLU$ activation function, which is widely used in feed forward neural networks due to its nonlinear characteristics, sparse activation, and efficient computing speed (Glorot, Bordes, and Bengio 2011). To enhance the efficiency and stability of model training, we normalized t and v prior to inputting it into the embedding layer.

To effectively integrate the observation values and their corresponding time information, the feature vectors obtained from both the time and value embedding processes were added together. This yielded a fused embedding vector (f) that served as the input for subsequent feature extraction stages. The structure of self-attention layer in feature extraction and fusion stage is identical to that of the default Transformer attention

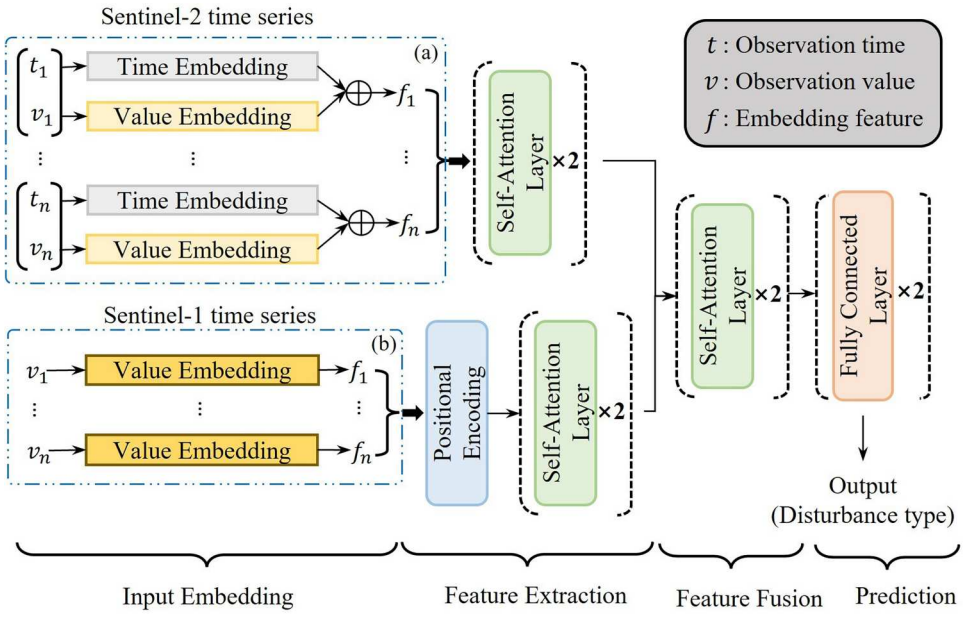


Figure 6. Architecture of IRMST model.

block. Unlike the standard learnable positional encoding, the IRMST model leveraged the specific DOY value and learnable parameters within the time embedding module. This design empowered the model to better adapt to forest disturbance detection tasks and handle the inherent irregularities present in remote sensing time series data.

(2) Sentinel-1 SAR time series data feature extraction

The Sentinel-1 time series data employed in this study consists of two bands (VV and VH), obtained through monthly mean compositing of all available images within each month. This process yielded a fixed annual time series length of 12. In contrast to Sentinel-2 multispectral data, Sentinel-1 SAR data exhibited less susceptibility to cloud cover. The consistent time interval and equal length across different years and regions for Sentinel-1 SAR data alleviated the need for an additional time information extraction module. However, to amplify the variations in backscatter coefficients caused by surface changes, the feature extraction module for SAR data also incorporated value embedding. We also normalized the backscatter coefficient data before inputting it to value embedding layer. The value embedding module for Sentinel-1 data shared the same structure as its counterpart for Sentinel-2 data (Figure 6b). However, their learnable parameters were independent, allowing the model to capture distinct feature representations for these two sensor types. Due to the consistent time intervals within the Sentinel-1 time series, the standard Transformer positional encoding could be directly employed to represent the positional relationships of elements within the sequence.

(3) Feature fusion and disturbance type classification

To integrate features extracted from Sentinel-1 and Sentinel-2 time series data, the IRMST model adopted a dual-branch structure with feature-level fusion (Figure 6). The model leveraged separate branches to process Sentinel-1 and Sentinel-2 data, and the extracted features were ultimately fused before forest disturbance type prediction. The embedding module conducted fundamental encoding operations on pixel-level Sentinel-1 and Sentinel-2 time series data, processing one year of time series at a time. Subsequently, the feature extraction module utilized two self-attention layers to extract features from the encoded Sentinel-1 and Sentinel-2 time series. Given the significant impact of self-attention layers on feature fusion, the feature fusion module incorporated two self-attention layers to comprehensively fuse temporal features

from multi-source time series. We experimented with various numbers of attention blocks (8, 6, 4, 2, 1) and observed no significant overfitting or underfitting when using two blocks. Consequently, we selected two attention blocks for the feature fusion layer. The disturbance type prediction module comprised two fully connected layers, facilitating judgments on the disturbance type (no disturbance, clear-cutting, disease, etc.) of the target pixel based on depth features.

2.3.2. Self-Supervised learning framework for IRMST

To address the issue of limited training data for forest disturbance types, this study proposed a self-supervised learning framework based on the IRMST model. This framework leveraged a large amount of unlabeled multi-source remote sensing time series data to learn more abstract and generalizable features. Similar to the data input to supervised IRMST model, the input data for self-supervised IRMST are also multi-source time series. We conducted spatially uniform sampling on all remote sensing data (two tiles, three years) with a step size of 16, and obtained a total of 2,823,576 multi-source time series samples without forest disturbance labels.

Self-supervised learning involved two key stages: unsupervised pre-training and supervised fine-tuning (Figure 7). During the unsupervised pre-training stage, the IRMST model's backbone network was trained using contrastive learning on unlabeled multi-source data. This process aimed to develop a robust feature extractor. The backbone network encompassed the IRMST model's multi-source data embedding module, feature extraction module, and feature fusion module. Contrastive learning focused on identifying common features within the dataset by comparing positive and negative samples. In essence, it aimed to bring the representations of similar samples (positive pairs) closer together in the feature space, while pushing the representations of dissimilar samples (negative pairs) further apart. As depicted in Figure 7, the input data (X) represented the unlabeled multi-source time series from Sentinel-1 and Sentinel-2 sensors. Positive samples ($X^{positive}$) for a given sample (X) were defined as time series data from different years but with the same spatial location. Conversely, negative samples ($X^{negative}$) represented time series data from any year but at a different spatial location compared to the original sample (X). The same network was utilized to extract features from $X^{positive}$, $X^{negative}$, and X, resulting in corresponding features ($Z^{positive}$, $Z^{negative}$, and Z, respectively). Finally, the model parameters were updated through a contrastive loss function and gradient backpropagation.

The supervised fine-tuning stage involved replacing the projection head with a classifier (the last two fully-connected layers in the IRMST model) to create a complete IRMST model incorporating the pre-trained backbone network. This complete IRMST model was then fine-tuned using labeled disturbance samples ($X_{labeled}$) for the final classification task.

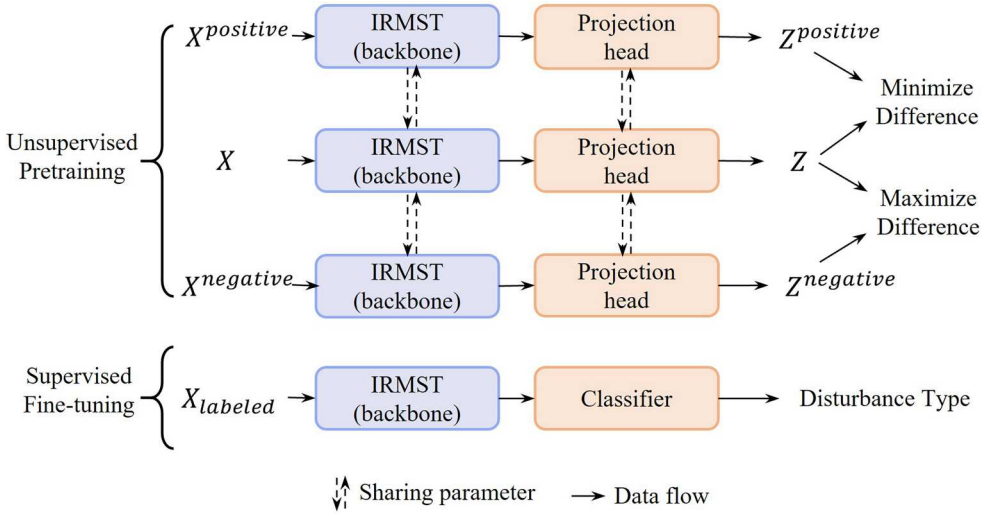


Figure 7. Self-supervised learning framework for IRMST.

2.4. Experiment setting

2.4.1. Effectiveness of the temporal feature extraction module

To assess the efficacy of the IRMST model's temporal feature extraction module for Sentinel-2 data, we conducted a comparative experiment evaluating other methods for processing temporal information. Building on previous studies of Transformer-based remote sensing time series models (Huang et al. 2023; Tarasiou, Chavez, and Zafeiriou 2023; Yuan and Lin 2021), we employed three additional approaches for temporal information extraction: (1) No DOY Information (ND): This baseline method entirely discarded temporal information during model training. (2) DOY as Additional Band (DaB): This approach incorporated DOY as an extra feature within the input data, effectively increasing the number of Sentinel-2 bands from 10 to 11 (denoted as *v*' for the modified input data). It's important to note that this method used standard modules of Transformer and did not involve designing a separate model structure specifically for processing temporal information (Vaswani et al. 2017). (3) DOY as Positional Encoding (DaPE): This method directly encoded DOY as positional encoding, leveraging this representation to capture the order of observations within the time series (without employing time embedding) (Zeng et al. 2023). A visual representation of methods DaB and DaPE was shown in Figure 8.

2.4.2. Effectiveness of multi-source data in the IRMST model

To evaluate the impact of multi-source data on disturbance type detection accuracy, we conducted ablation experiments using the IRMST model. We trained and tested the model under three data source configurations: (1) Sentinel-1 Time Series Only: This scheme utilized solely Sentinel-1 data for training and testing. The IRMST model structure was adjusted accordingly by removing the redundant data embedding and feature extraction modules for Sentinel-2 data. Specifically, Sentinel-2 data embedding module and the two self-attention layers dedicated to its feature extraction were removed. (2) Sentinel-2 Time Series Only: This scheme employed only Sentinel-2 data for training and testing. Similar to the previous scenario, the IRMST model structure was adapted by removing unnecessary components for processing Sentinel-1 data. (3) Combination of Sentinel-1 and Sentinel-2 Time Series: This configuration utilized both Sentinel-1 and Sentinel-2 data for a comprehensive analysis. The complete IRMST model structure, as described earlier, was employed in this setting.

2.4.3. Improvement of self-supervised learning

To quantitatively assess the effectiveness of self-supervised learning for forest disturbance type detection, we conducted the following comparative experiments: (1) IRMST with supervised learning: The IRMST model was directly trained on the labeled disturbance samples without any pre-training stage. This served as the baseline for comparison. (2) IRMST with self-supervised learning: The IRMST model was pre-trained using the self-supervised learning framework described above. This pre-trained model was then fine-tuned with

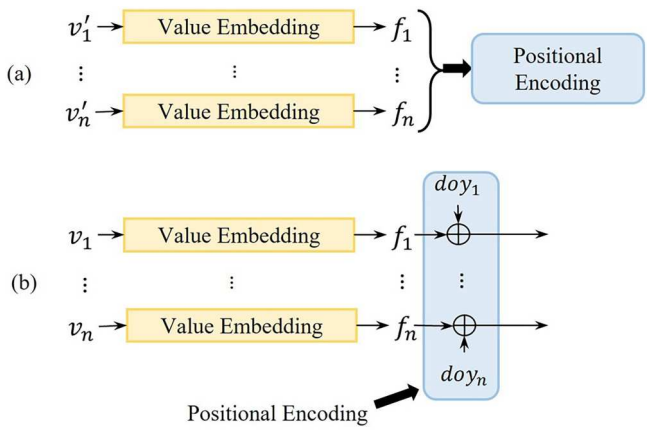


Figure 8. Temporal feature extraction methods for comparison. (a) DOY as an additional Sentinel-2 band (DaB); (b) DOY as positional encoding (DaPE).

labeled disturbance samples for forest disturbance type classification. The performance of both models (self-supervised and supervised) was evaluated and compared in terms of forest disturbance type detection accuracy. (3) Impact of sample size on self-supervised and supervised learning: We divided the available training data into five subsets representing different sample sizes: 20%, 40%, 60%, 80%, and 100% of the total training data. For each sample size level, we trained both the self-supervised learning and supervised learning models using the corresponding subset of training data. The performance of both models was then evaluated on a held-out test set to assess the impact of sample size on detection accuracy. This experiment allowed us to investigate the benefits of self-supervised learning for improving the generalizability of the IRMST model, particularly in scenarios with limited labeled training data.

2.4.4. Training details

For supervised training of the IRMST model, the batch size was 128. We employed the Adam optimizer, with parameters $\beta_1 = 0.9$ and $\beta_2 = 0.999$. The initial learning rate was set to 1e-5. The Focal Loss function (Lin et al. 2017) was used to measure the discrepancy between the predicted and actual class labels. Training was conducted for 50 epochs, with early stopping implemented to halt the process if the validation loss did not improve for 5 consecutive epochs. The model training was performed on an NVIDIA Quadro RTX 4000 GPU with 8GB of memory. The self-supervised learning framework included model pre-training and fine-tuning, with the fine-tuning process using parameters identical to those used in supervised learning framework. During pre-training, the batch size was 512, the learning rate was 1e-4, and we used a contrastive loss function (Chen et al. 2020). All other parameters were also consistent with the supervised learning configuration.

2.4.5. Accuracy evaluation

Forest disturbance type detection accuracy was evaluated using a dataset of 310 disturbance samples represented as polygons. It is important to note that the IRMST model predicted the disturbance type for individual pixels. Therefore, for each test sample (polygon), a prediction was considered correct only when the number of correctly classified pixels within the polygon exceeded half of its total area (Linke et al. 2017). We employed User's Accuracy (UA), Producer's Accuracy (PA), and F1 score to assess the model's classification performance for each disturbance category (Zhong, Hu, and Zhou 2019). UA indicates the probability that a pixel classified by the model as belonging to a specific disturbance category actually belongs to that category. PA represents the probability that a pixel belonging to a particular disturbance category was correctly classified by the model into that category. F1 score is the harmonic mean of UA and PA, providing a balanced view of model performance. It is particularly useful for evaluating models on imbalanced datasets, which are common in remote sensing applications where certain disturbance types might be less frequent (Zhong, Hu, and Zhou 2019). The overall F1 score of the model was calculated as the average F1 score across all disturbance categories. The formula for calculating the overall F1 score was provided below:

$$F1 = \frac{1}{N}\sum_{i=1}^{N} F1_i$$

$$F1_i = \frac{2{*}UA_i{*}PA_i}{UA_i + PA_i}$$

Where N is the number of disturbance types, and i is the ith disturbance type.

3. Results

3.1. Performance of IRMST on forest disturbance types detection

3.1.1. Evaluation of temporal feature extraction for Sentinel-2 time series

The IRMST model achieved the highest F1 score for each disturbance type, surpassing other methods by a margin of 5% to 14% (Table 2). Notably, the DaPE method exhibited the lowest accuracy across all disturbance types. This suggested that directly replacing positional encoding with raw time information

Table 2. Accuracy assessment of different methods on temporal feature extraction.

	ND			DaB			DaPE			IRMST		
Disturbance type	UA	PA	F1	UA	PA	F1	UA	PA	F1	UA	PA	F1
No disturbance	0.96	0.93	0.95	0.95	0.91	0.93	0.96	0.84	0.89	0.98	0.95	0.97
Clear-cutting	0.95	0.98	0.96	0.93	0.97	0.95	0.91	0.97	0.94	0.95	1.00	0.97
Disease	0.78	0.78	0.78	0.74	0.74	0.74	0.50	0.78	0.61	0.95	0.87	0.91
Overall F1	0.90			0.87			0.81			0.95		

(DOY) was not an effective approach for extracting temporal features. Interestingly, the model without any DOY information (ND) outperformed both DaB and DaPE. This indicated that omitting temporal information altogether could be more effective than using inappropriate methods for extracting temporal features.

To further illustrate the differences in temporal feature extraction between the four strategies, a representative clear-cutting patch was analyzed as an example. Figure 9 presents the multi-band time series for this patch alongside the attention maps generated by each method. There was a clear decrease in reflectance across multiple bands around the 72nd time step (Figure 9a), signifying the clear-cutting event near the end of the year. The ND method distributed attention weights across various time steps, lacking specific focus on the clear-cutting event (Figure 9b1). The DaB method assigned some attention to the time of clear-cutting, but it dispersed attention weights across other time points (such as around the 50th time step), suggesting less focused disturbance event (Figure 9b2). The DaPE method exhibited more scattered attention weights, suggesting an inability to learn effective temporal features (Figure 9b3). The IRMST model predominantly focused on data near the 72nd time step, with the highest attention weight concentrated around the 72nd (Figure 9b4). This indicated successful identification of the clear-cutting event. It's important to note that all three methods (DaB, DaPE, and IRMST) assigned high attention weights to positions after the 76th time step. This arose from a pre-defined input length of 100 for the time series. Sentinel-2 time series data exhibited varying lengths, and zero padding was applied to those shorter than 100. Consequently, the data after the 76th time step held no meaningful information, yet these models allocated attention to this padded region. Notably, the IRMST model exhibited a reduced focus on this area compared to DaB and DaPE, suggesting a more robust temporal feature extraction module that mitigated this issue. These observations reinforced the effectiveness of the IRMST model's temporal feature extraction capabilities. It successfully identified the clear-cutting event and demonstrated a more targeted focus on informative time steps compared to the other methods.

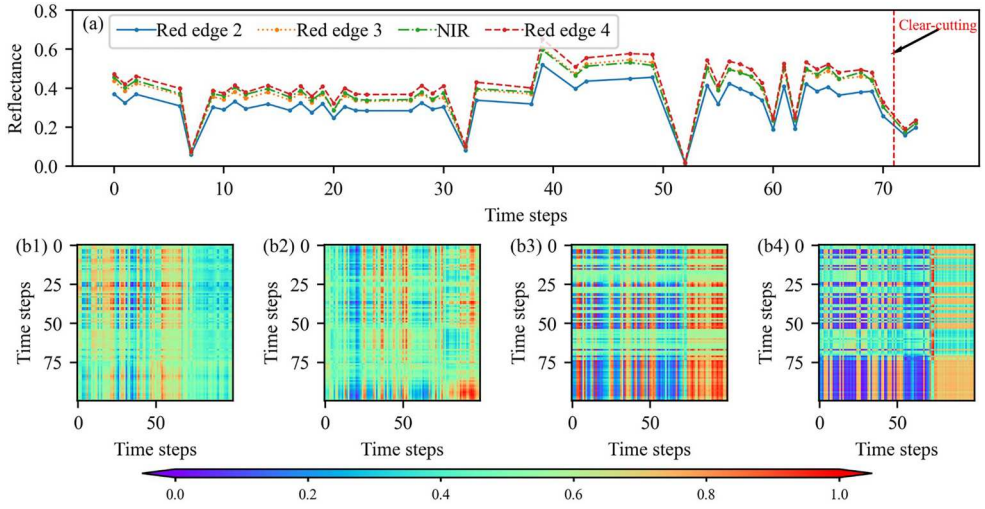


Figure 9. Multi-band time series of a typical clear-cutting sample and attention maps of different models. (a) Sentinel-2 time series of the typical clear-cutting sample. (b1) Attention map of ND. (b2) Attention map of DaB. (b3) Attention map of DaPE. (b4) Attention map of IRMST.

3.1.2. Impact of multi-source data on disturbance type detection

The influence of multi-source data on disturbance type detection accuracy was evaluated using the IRMST model (training with supervised learning) with different data configurations (Table 3). Employing multi-source data consistently improved IRMST model performance compared to using single-source data (Sentinel-1 or Sentinel-2). Utilizing only Sentinel-1 time series data at the pixel level resulted in poor performance. The model trained solely on Sentinel-1 data exhibited a F1 score of just 0.59, indicating its ineffectiveness for forest disturbance type detection. Adding Sentinel-1 data to Sentinel-2 data yielded some improvement, particularly for forest disease detection. However, it led to a slight decline in clear-cutting detection compared to using Sentinel-2 data alone.

3.2. Impact of self-supervised learning

3.2.1. Improvement of disturbance type detection accuracy

Table 4 summarizes the accuracy metrics achieved by the self-supervised and supervised IRMST models for each disturbance type. The F1 score obtained through self-supervised learning surpassed that achieved solely through supervised learning. Specifically, self-supervised learning enhanced the accuracy of the IRMST model in detecting instances of no-disturbance and clear-cutting. The accuracy for detecting disease remained comparable to that achieved through supervised learning. To delve deeper into this phenomenon, a representative no-disturbance sample and the attention distributions from both models were analyzed through Figure 10. Notably, the self-supervised model's focus points (DOY 194 and 289) coincided with cloud-free conditions observed in the images. In contrast, the supervised model focused on time steps with thin clouds (DOY 89 and 344). It suggested that self-supervised learning could effectively mitigate the influence of thin clouds on disturbance detection. The supervised model, on the other hand, might have been misled by subtle changes in spectral data caused by thin clouds, resulting in misclassifications.

3.2.2. Impact of sample size on self-supervised learning and supervised learning

Figure 11 depicts the overall F1 score of the IRMST models under different sample size and training methods (self-supervised and supervised learning). When dealing with limited training data (20% of all samples), the self-supervised IRMST model achieved a significantly higher accuracy (0.95) compared to the supervised model (0.86). This demonstrated the ability of self-supervised learning to mitigate the negative impact of small sample sizes. As the sample size increased, the accuracy of the supervised learning model gradually improved. By the time the sample size reached 60%, its accuracy approached that of the self-supervised model. Furthermore, the self-supervised model's accuracy remained stable around 0.95 with increasing sample size, indicating its convergence in small sample scenarios. Even with larger sample sizes (60% to 100%), the self-supervised learning model exhibited slightly higher accuracy compared to the supervised model. This suggested that self-supervised learning not only alleviated small sample issues but potentially improved the upper limit of achievable model accuracy.

Table 3. Accuracy assessment of different data sources.

	Sentinel-1			Sentinel-2			Sentinel-1 + Sentinel-2		
Disturbance type	UA	PA	F1	UA	PA	F1	UA	PA	F1
No disturbance	0.75	0.98	0.85	0.98	0.96	0.97	0.98	0.95	0.97
Clear-cutting	0.96	0.75	0.85	0.97	0.99	0.98	0.95	1.00	0.97
Disease	1.00	0.04	0.08	0.87	0.87	0.87	0.95	0.87	0.91
Overall F1	0.59			0.94			0.95		

Table 4. Accuracy assessment of self-supervised IRMST and supervised IRMST.

	Self-supervised IRMST			Supervised IRMST		
Type	UA	PA	F1	UA	PA	F1
No disturbance	0.98	0.98	0.98	0.98	0.95	0.97
Clear-cutting	0.99	1.00	0.99	0.95	1.00	0.97
Disease	0.95	0.87	0.91	0.95	0.87	0.91
Overall F1	0.96			0.95		

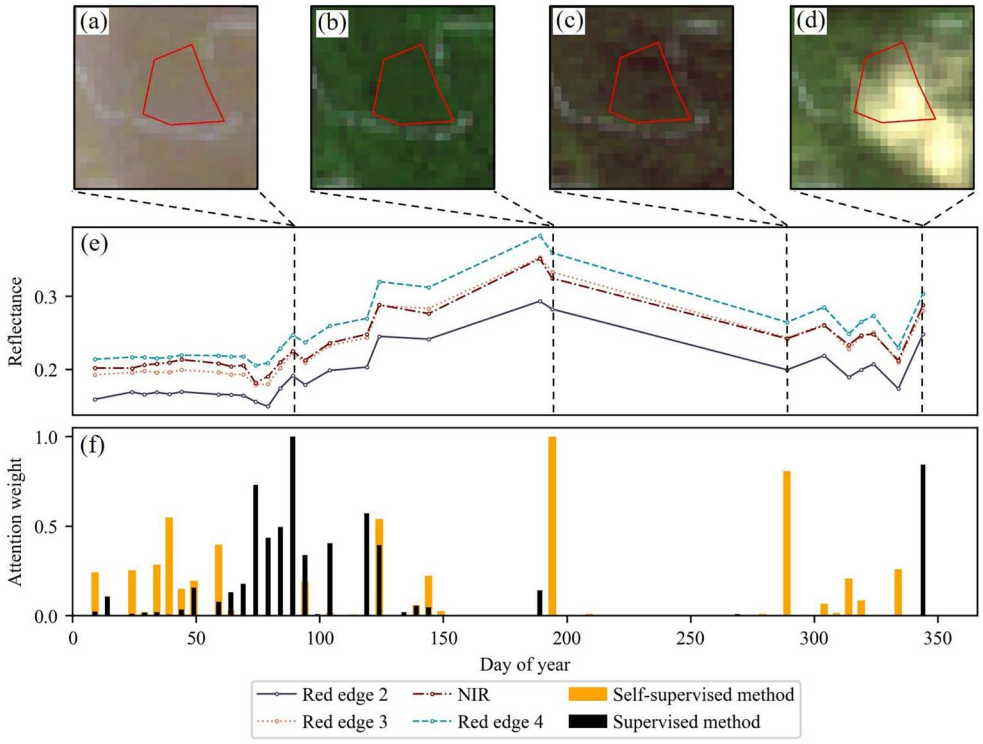


Figure 10. Impact of self-supervised learning on mitigating cloud effects in disturbance detection. (a-d) True-color Sentinel-2 images of a no-disturbance sample at four time steps: (a) DOY 89 (supervised learning focus point) – Thin clouds. (b) DOY 194 (self-supervised learning focus point) – Cloud-free. (c) DOY 289 (self-supervised learning focus point) – Cloud-free. (d) DOY 344 (supervised learning focus point) – Thin clouds. (e) Sentinel-2 time series reflectance of the no-disturbance sample. (f) Attention weight distribution of self-supervised and supervised learning models on the no-disturbance time series. The polygons in images represent the bounding of the selected no-disturbance sample.

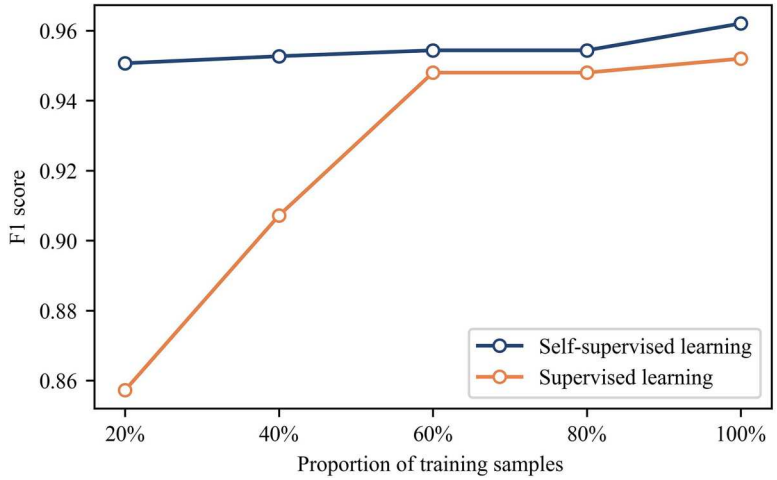


Figure 11. Impact of sample size on IRMST model accuracy with self-supervised and supervised learning.

3.3. Spatiotemporal distribution of forest disturbance types

The IRMST model successfully generated a spatiotemporal distribution map of forest disturbance types across the study area (Figure 12). This map revealed significant variations in the spatial distribution of clear-cutting events over the investigated years (2020-2022). Clear-cutting activities were primarily concentrated in the central and southwestern regions of the study area in 2020, and shifted to the northwestern and

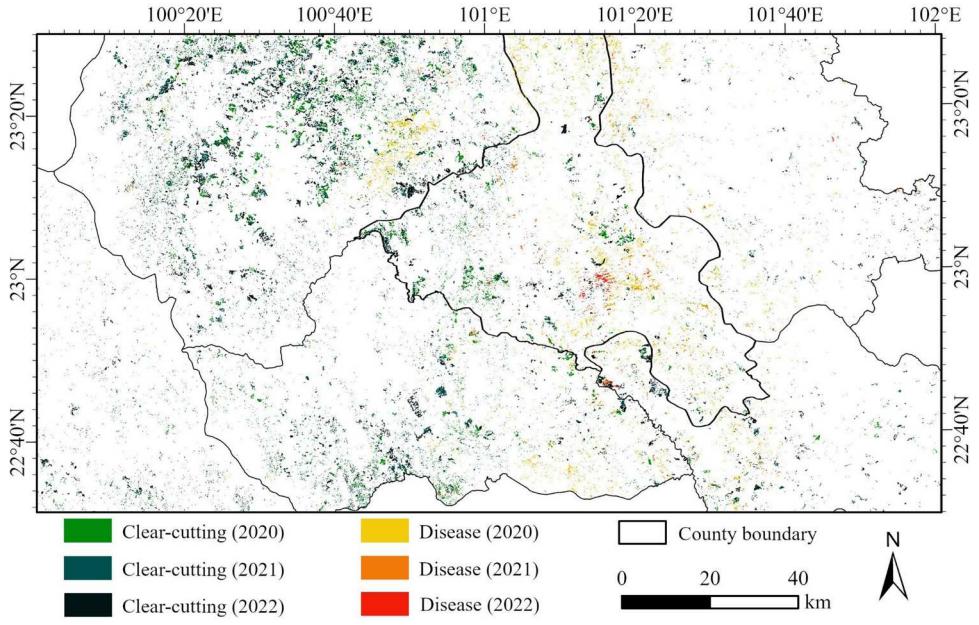


Figure 12. Spatiotemporal distribution map of disturbance types.

Table 5. Forest disturbance area (km²) between 2020 and 2022.

Year	Clear-cutting	Disease	Total
2020	306.09	47.56	353.65
2021	240.97	18.47	259.44
2022	288.71	24.16	312.87
Total	835.77	90.19	925.96

southwestern regions in 2021 and 2022. The map highlighted a more persistent and severe disease disturbance concentrated in the central part of the study area. Additionally, a notable presence of disease disturbance was observed in the northern region during 2020.

Table 5 summarized the total forest disturbance area for each year between 2020 and 2022. The total area affected by disturbances over this period was approximately 925.96 km². No statistically significant trend in the overall forest disturbance area was observed across the three years. The disturbance area fluctuated slightly year-to-year, with the largest area (353.65 km²) occurred in 2020 and the smallest (259.44 km²) in 2021. The area affected by disease disturbance consistently remained lower than the area affected by clear-cutting across all three years. Similar to the overall trend, the area of both clear-cutting and disease disturbance peaked in 2020 (306.09 km² for clear-cutting and 47.56 km² for disease) and reached a minimum in 2021 (240.97 km² for clear-cutting and 18.47 km² for disease).

4. Discussions

4.1. Representation of time information in deep learning

Our study highlights the importance of accurately representing time information in forest disturbance detection algorithms, particularly in the context of irregular time series data affected by cloud cover and other weather conditions. We adopted a time feature fusion approach, which integrates time information with other features to improve model performance. Our study innovatively implemented a dedicated module specifically designed to extract temporal features based on Transformer. It maps individual observation times to positional encoding vectors, with the mapping functions being learnable rather than predefined. Transformer-based architectures for time series analysis typically focus on the design of positional encoding (Wen et al. 2023). Unlike gated RNNs, these models generally don't utilize time interval data as direct input (Tarasiou, Chavez, and Zafeiriou 2023). This might be attributed to the underlying principles of attention

mechanisms employed in Transformers. Their ability to simultaneously focus on global features across different time steps necessitates absolute time information, such as timestamps. Tarasiou, Chavez, and Zafeiriou (2023) addressed this by utilizing a learnable lookup table to construct a global mapping between observation times and positional encoding. However, introducing an additional lookup table can increase the operational complexity of the model, potentially hindering its practical application. Similar to the idea of a lookup table, this study also established a mapping from timestamps to feature vectors. However, this structure is embedded in the IRMST network and can be directly trained end-to-end without the need for additional lookup tables.

The findings of this study emphasize the crucial role of accurate temporal information representation in time series analysis for forest disturbance detection. Inappropriate methods for extracting temporal features can potentially hinder the model's ability to exploit the inherent temporal patterns within the data. In some cases, as our results suggest, omitting temporal information altogether might lead to better performance compared to using an ineffective feature extraction method. Inappropriate temporal feature extraction methods might misinterpret the irregular distribution of observations caused by missing data as actual temporal patterns. This can lead the model to focus on irrelevant information and hinder its ability to capture the true underlying temporal dynamics associated with forest disturbances. In addition, introducing complex time feature extraction modules can increase the overall complexity of the model. This can lead to overfitting, especially when dealing with limited training data. Furthermore, these complex modules might not be well-suited for capturing the specific temporal characteristics relevant to forest disturbance detection in the presence of missing data. Investigating hybrid approaches that combine strengths from both weighted and time feature fusion methods, and exploring alternative methods for incorporating time information into Transformer-based architectures could be promising direction for future research. By focusing on appropriate temporal feature extraction methods, future research can potentially lead to the development of more robust and generalizable models for forest disturbance detection using remote sensing data, even in scenarios where data is incomplete due to cloud cover or other interfering factors.

The performance of time series feature extraction is also related to the degree of data preprocessing. The data acquisition and preprocessing workflow in this study is relatively streamlined. We neither deliberately selected equal-interval sequences nor performed imputation on missing values. Instead, we refined our model architecture to better address these challenges. Conventional approaches typically selected a fixed number of images each year to construct datasets and then impute missing values. For example, Marinelli et al. (2023) generated equal-interval time series through monthly compositing. Although this strategy simplified subsequent algorithm design, imputation inevitably introduced additional errors and may have obscured genuine forest disturbance signals. In our study, to fully leverage all available remote-sensing information, we analyzed nearly every accessible Sentinel-2 image over the course of a year. For missing pixels, we employed the 'zero-filling' method. Though this approach may have attenuated disturbance signals, it preserves the timestamp associated with each image. Even if pixels were missing, the corresponding temporal label enabled the model to recognize missing-data patterns. Masking missing values could also reduce the negative impact on disturbance detection, but it disrupted temporal continuity.

4.2. The role of self-supervised learning

High-performance deep learning models often require vast amounts of labeled training data. Unfortunately, acquiring labeled forest disturbance samples is a time-consuming and laborious process, leading to a scarcity of suitable data. Self-supervised learning presents a promising solution by enabling the utilization of unlabeled remote sensing data from the target domain for pretraining deep learning models. This approach helps extract effective features without the need for extensive manual labeling. Current research on contrastive learning in remote sensing primarily focuses on spatial and spectral augmentations (e.g. random cropping, Gaussian noise) (Hou et al. 2022). There is a relative lack of exploration regarding temporal dimension augmentation (Wang et al. 2022). Xu, Ma, and Zhang (2024) proposed a self-supervised time series crop mapping method based on Transformers and contrastive learning, demonstrating its potential to alleviate sample scarcity. Our study used contrastive learning to pretrain IRMST network and treated time series data from different years but the same spatial location as positive pairs. This strategy aimed to train the model to minimize the influence of normal interannual dynamic changes (e.g. phenology, weather) on feature

extraction. This approach aligns with the notion that a universal strategy for constructing positive and negative samples can improve the performance of deep learning models in remote sensing analysis (Zhu et al. 2022). However, it is crucial to acknowledge that different remote sensing applications (e.g. single scene vs. time series analysis, vegetation vs. hydrology) necessitate task-specific strategies for optimal positive and negative sample pair definitions.

Beyond addressing sample scarcity, self-supervised learning offers additional advantages for the IRMST model. Similar to Xu, Ma, and Zhang (2024), our study found that self-supervised pretraining can mitigate misclassification errors and reduce the impact of cloud pollution pixels on disturbance detection. This improvement in accuracy can be attributed to several factors, including the increased representativeness of training samples, expansion of data volume and diversity, and the use of learnable positional encoding in the IRMST model. Overall, self-supervised learning holds significant promise for improving the performance of deep learning models in remote sensing applications, including forest disturbance detection.

4.3. The role of SAR data in disturbance detection

The role of SAR data in forest disturbance detection is significant, particularly in regions where optical data is heavily impacted by cloud coverage. SAR data offers the advantage of being less affected by clouds, making it a valuable resource for monitoring forest disturbances. However, SAR data also presents challenges, such as strong noise that can cause small-scale or weaker forest disturbances to be overlooked (Sun et al. 2023). Previous research has indicated that pixel-level features of single-scene SAR images have limited effectiveness in detecting deforestation (Hethcoat et al. 2021). Our study further corroborates this finding by demonstrating that using pixel-level Sentinel-1 time series data alone results in poor detection performance for forest disturbance types. We also found the detection accuracy of clear-cutting is higher than for diseases. This result is consistent with the response of Sentinel-1 to different forest disturbance types, namely, Sentinel-1 has a response to clear-cutting (but the boundary is unclear) and the response to diseases is very weak.

When fusing Sentinel-1 SAR data with Sentinel-2 multispectral time series, the accuracy of detecting clear-cutting disturbances decreased, while the accuracy of detecting diseases improved compared to using only Sentinel-2 data. The potential reason for the limited effectiveness of SAR data in detecting clear-cutting could be the lack of spatial feature utilization in our proposed method. Clear-cutting events often lead to drastic changes in surface spatial texture, which SAR data can respond to in visual features of images. However, without spatial information, SAR data may struggle to accurately detect such disturbances. This is further supported by the work of Hethcoat et al. (2021), who achieved better results by incorporating texture analysis from SAR data for high-intensity selective logging detection. Therefore, when spatial information is disregarded, incorporating SAR data tends to weaken the representativeness of clear-cutting features extracted from Sentinel-2 time series. Conversely, disease disturbances typically result in weaker changes in forest spatial texture, but tend to lead to changes in vegetation moisture content (Cotrozzi 2022). Therefore, a single Sentinel-1 pixel may be beneficial for characterizing disease disturbances, explaining why the addition of Sentinel-1 pixel-level time series improved the detection accuracy of disease disturbances in our study. However, using only single pixel SAR time series without multispectral information cannot effectively detect clear-cutting. Extracting and leveraging spatial information from SAR data is crucial for improving complex disturbance type detection. Deep learning methods can potentially automate this process, but their effectiveness depends on the availability of high-quality training data. Self-supervised learning can potentially address challenges related to sample quality for deep learning models using SAR data. By relying on self-supervised algorithms and proxy tasks, this approach can potentially reduce the training difficulty for spatial feature extractors specifically designed for SAR data.

Improving the performance of SAR data at the algorithm level was crucial, but enhancements at the data level should not be overlooked either. Different types of SAR data contributed differently to forest disturbance detection. Radar wavelength played a key role: for instance, Dalponte et al. (2023) found that COSMO-SkyMed (X-band) outperformed Sentinel-1 (C-band) in detecting windthrow events. While C-band is more sensitive to logging-induced changes, X-band, with its shorter wavelength, captured finer variations in the canopy surface, making it more responsive to subtle or superficial disturbances. Fusing both types of data could potentially enhance detection accuracy. However, the framework in Dalponte et al. (2023) struggled

with integrating SAR data of different spatial and temporal resolutions. In contrast, deep learning architectures, equipped with modules like convolution and self-attention, allow for efficient feature-level fusion of heterogeneous SAR data.

Despite these advantages, SAR-based disturbance detection continues to be challenged by terrain effects. Topography-induced distortions significantly reduced SAR performance. Terrain correction using a digital elevation model (DEM) was typically required to mitigate pseudo-changes caused by slope and aspect. Moreover, polarimetric decomposition methods such as Cloude–Pottier (Cloude and Pottier 2002) could separate radar backscatter into physically meaningful scattering mechanisms (e.g. volume, surface, and double-bounce scattering). Leveraging these decomposed components helps distinguish real canopy changes (caused by disturbances like clear-cutting or insect outbreak) from variations due to terrain, thereby reducing background noise and improving detection reliability.

4.4. Transferability of the model

Due to significant variations in spectral characteristics and disturbance manifestations across different regions and time periods, models trained on limited spatial extents or small datasets often experience substantial declines in detection accuracy when applied to broader, more heterogeneous datasets. When transferring the IRMST model to other regions, we must account for differences in disturbance types, environmental conditions, and data availability. Disturbance events such as logging, insect outbreaks, and windthrows differ in spatial scale, intensity, and temporal progression, which can hinder the generalizability of learned features. This study considered both clear-cutting and disease, representing abrupt and gradual disturbances, respectively. The IRMST model performed well in both aspects. This suggests that the model has the potential to generalize to other abrupt or gradual disturbance types, provided it undergoes modest fine-tuning.

Beyond disturbance types, natural environmental differences (e.g. vegetation composition and seasonal dynamics) can also affect model transferability. These factors may cause confusion between natural phenological changes and actual disturbance signals. Since our study area is located in a subtropical region characterized by evergreen vegetation and an absence of snow, transferring the model to temperate zones would require particular attention to the effects of deciduous leaf-off periods and snow cover. In such cases, additional disturbance samples may be necessary to achieve satisfactory detection performance.

4.5. Limitations and future work

Detecting forest disturbance accurately requires precise information regarding both the timing and types of disturbances. However, obtaining highly specific and accurate disturbance time samples can be challenging. In this study, due to the difficulty of acquiring such precise time samples, we did not further subdivide the samples within each year. Nevertheless, our model demonstrates the capability to capture temporal information effectively by focusing on features corresponding to specific times. Moreover, while our study focused on detecting clear-cutting and disease disturbance types, it did not distinguish between insect-induced disturbances and diseases. Recognizing the potential differences between these types and their respective occurrence times is crucial for comprehensive disturbance detection. The limited availability of samples further constrained our ability to explore additional disturbance types.

Given that some forest disturbance events were relatively rare, the sample classes became highly imbalanced, causing the model to overfit to the minority class. To mitigate this issue, we introduced a focal loss. Additionally, during data preprocessing, oversampling techniques can be employed to augment the minority class. For instance, random replication or the SMOTE method (Chawla et al. 2002), which generates new minority-class samples by interpolating within the feature space can increase the sample size. Although down-sampling methods that reduce the number of majority-class samples randomly or heuristically can also address class imbalance, they were not appropriate for our study, as they would result in a significant shrinkage of the overall sample size. On the other hand, the self-supervised learning approach used here leveraged large amounts of unlabeled data to pretrain the model, thereby further enhancing its robustness and generalization capability.

The configuration of hyperparameters (e.g. learning rate, network depth, batch size) has a profound impact on model performance. Improper settings could not only hinder the model's convergence but also lead to overfitting or underfitting. To enhance the generalization of the IRMST model, this study employed cross-validation for evaluation. Our approach also has room for improvement. By incorporating systematic hyperparameter search methods such as Bayesian optimization, random search, or grid search, we can further refine our experimental plan and reduce the model's sensitivity to hyperparameter settings. Moreover, integrating regularization techniques such as dropout and batch normalization can further enhance the stability of model convergence.

With the advancement of remote sensing technology, various types of data have been applied to forest monitoring. For instance, Light Detection and Ranging (LiDAR) technology provides direct information on canopy height, forest structure, and vertical stratification. Meanwhile, the sub-meter spatial resolution of high-resolution Unmanned Aerial Vehicle (UAV) imagery enables the extraction of detailed surface textures and canopy gap characteristics. Incorporating a multi-scale feature pyramid network to upgrade the IRMST model can facilitate the fusion of texture and spectral features across different spatial resolutions, thereby significantly enhancing the model's ability to capture spatial patterns and relationships.

Future research may aim to enhance the model's capability to detect a wider range of forest disturbances. Our self-supervised IRMST model can be readily applied and tested for detecting additional disturbance types, thus offering a promising avenue for further research. By incorporating a more diverse dataset of forest disturbances, the approach presented in this study can be extended across a broader spatiotemporal domain. It can also elevate the temporal resolution of disturbance detection to near the revisit frequency of remote sensing satellites. This not only optimizes forest resource management and enhances economic returns, but also enables timely identification and prevention of large-scale ecological degradation.

5. Conclusion

Forests in subtropical regions are susceptible to various disturbances like logging and diseases. These disturbances significantly impacted the health and functionality of forest ecosystems. Accurate identification of the types and timing of these disturbances is crucial for maintaining forest ecological security and for informing decisions related to human societies that depend on these ecosystems. This study addressed this challenge by proposing a deep learning model, IRMST, for forest disturbance detection using multi-source remote sensing time series data. The IRMST model incorporated two key advancements: (1) Improved time information encoding: The Transformer architecture within IRMST utilized an enhanced method for encoding temporal information from the time series data. This improvement facilitated more accurate extraction of temporal features from Sentinel-2 data, leading to better disturbance detection accuracy. Our findings also emphasized the importance of appropriate time information encoding, as ineffective methods can hinder model performance. (2) Self-supervised learning for robustness: The IRMST model leveraged self-supervised learning for pretraining. This approach addressed the scarcity of labeled data for forest disturbance detection and enhanced the model's overall accuracy and stability. Self-supervised learning also improved the model's ability to handle noise in the data, such as thin clouds. Experiments confirmed that self-supervised learning offered superior generalizability, particularly in situations with limited training data, compared to supervised learning methods alone.

The IRMST model offers a promising approach for accurate forest disturbance detection in subtropical regions using multi-source remote sensing time series data. In addition, we found that while pixel-level Sentinel-1 data exhibited a less significant response to clear-cutting, it showed a more pronounced response to diseases. This underscored the importance of utilizing spatial information in SAR data for effective disturbance detection. The findings of this study have significant implications for addressing problems related to feature extraction from irregular time series and limited disturbance type samples.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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Data availability statement

The data that support the findings of this study are available on request from the corresponding author.

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